

Multi-Party Functional Encryption (MPFE): A tool in the distributed and decentralized world

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Abstract

Functional Encryption (FE) is a concept that generalizes public-key encryption, allowing a party that owns a private key to find a function of the plaintext (instead of the plaintext itself). Multi-Party Functional Encryption (MPFE) generalizes this concept and adapts it to multi-party settings, allowing for decentralization in both the ciphertexts—which might originate from multiple sources—and the keys—thereby eliminating the necessity of a central authority and avoiding the introduction of a single point of trust and failure. The current paper presents a substantial foundation of MPFE to the non-specialist reader. It provides definitions, classifications, and discusses properties of MPFE and its relation with other cryptographic concepts. The potential applicability of MPFE, which covers multiple domains and use cases, is discussed. The paper investigates the real-world adoption of MPFE, including its presence in technical specifications or its availability in open-source libraries. Finally, the current study discusses challenges and therefore opens up new research directions.

Keywords. Functional encryption, multi-input functional encryption, multi-party functional encryption, identity-based encryption, attribute-based encryption.

1 Introduction

Public-Key Cryptography (PKC) revolutionized the digital world since its introduction 50 years ago [1]. Its applicability is vast, enabling the existence of

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the internet and its use in protocols such as TLS. Nevertheless, Public Key Encryption (PKE) ensures access to data in the *all-or-nothing* setting: one either completely decrypts the ciphertext or learns nothing (up to a given negligible probability and except for a bound on its length) about the plaintext [2]. Functional Encryption (FE) is a concept that overcomes this limitation, enabling partial knowledge of the plaintext. More precisely, in FE, a decryption key allows a user to learn a given function of the encrypted data, without necessarily revealing the full plaintext. A more general concept, Multi-Input Functional Encryption (MIFE) allows a user to learn a function on n plaintexts given the corresponding ciphertexts, each encrypted under a different public key [3]. Even more, Multi-Party Functional Encryption (MPFE) aims to unify all flavors of FE, supporting decentralization across both ciphertexts and decryption keys. While simple FE is well-suited for data from a single source, MPFE is suitable for applications in which the input originates from multiple parties. MPFE can eliminate the need for centralized authority, thereby avoiding the introduction of a single point of trust and failure [4].

1.1 Motivation

FE is a hot research topic nowadays, but there is still room for improvement, especially in practical usability. Last year, at CRYPTO 2025, EUROCRYPT, and PKC 2025, three of the most prestigious conferences on cryptography, more than 15 papers on FE were presented. The potential applicability of MPFE is extensive, covering multiple domains and use-cases, including, among others, delegation of computation, secure data mining, federated analytics, Machine Learning (ML), and secure communication. However, compared to similar techniques, mostly adopted in niche areas, MPFE still requires further work to reach a level of maturity that allows its implementation in production. Our findings indicate that MPFE is primarily academic at this stage: in its general form, it is not adopted in open-source or production-ready libraries and industry standards, which instead refer to specific, better-established concepts, such as ABE and IBE. Moreover, even in the academic world, the MPFE lacks maturity due to its novelty, in the absence of the test of time.

1.2 Aim and contribution

We aim to provide a basic understanding of MPFE to the non-specialist reader. We consider this of great importance, as much of the existing work is very technical and difficult for starters. We review the most popular concepts, recall their classification, and discuss the fundamental properties of MPFE. Although we allocate a significant space to FE, our focus is on MPFE in all its forms, including MIFE and multi-authorities FE. We are primarily interested in specific properties of MPFE, such as decentralization and dynamicity. We then focus on the use of MPFE across different fields, demonstrating its broad applicability. To summarize, the primary contributions of the paper are:

- We present a basic, mostly informal, but substantial foundation on MPFE to the non-specialist reader;
- We provide a review of the definitions, classifications, and properties of MPFE, discussing the relation with other cryptographic concepts;
- We give multiple use-case scenarios where MPFE finds its applicability, spread across different domains (e.g., networking, delegation of computation, data analytics);
- We analyze the real-world adoption of MPFE, discussing aspects such as its presence in technical specifications or its availability in open-source libraries;
- We discuss challenges and therefore open up towards new research directions.

1.3 Novelty with respect to related works

We further place our work in relation to similar existing works. The paper [5] is a functional encryption survey that provides a taxonomy of FE schemes, presents the state of the art, and clearly lays out the mathematical background for understanding the constructions of the schemes presented. Contrary to our current work, the paper does not focus on MPFE or its applicability, and it is rather very technical. The survey [6] reviews FE and related primitives and explains how these schemes are used across IoT-related domains. The paper discusses decentralization and dynamicity briefly and does not target the multi-party case or domains outside IoT. Other existing works narrow the focus. Examples include [7], which explores ML using FE, and [8], which discusses secure computation using FE. The current work brings novelty compared to the previously mentioned works, which are already a few years old now. The domain of MPFE is highly dynamic. To support our claim, we identified 19 papers on ScienceDirect that match "Multi-Input Functional Encryption" OR "Multi-Client Functional Encryption" after 2022, 14 papers on IEEE Xplore, 61 papers on Springer Link, and 26 papers on IACR ePrint. We intentionally avoid any reference to reviews and surveys dedicated to ABE or IBE. For these, the reader can refer to [5,6]. Finally, we are unaware of any other paper that investigates the presence of MPFE in technical specifications or the current status of general-purpose MPFE implementations.

1.4 Outline

The remainder of the paper is organized as follows. Section 2 gives the necessary preliminaries. Section 3 presents the history and development of MPFE, indicating the timeline of the fundamental results. Section 4 defines ABE, IBE, FE, and MIFE. Section 5 defines MPFE and provides a unified taxonomy. Section 6 discusses the properties of interest. Section 7 investigates the relationship between MPFE and other related cryptographic concepts. Section 8 presents the

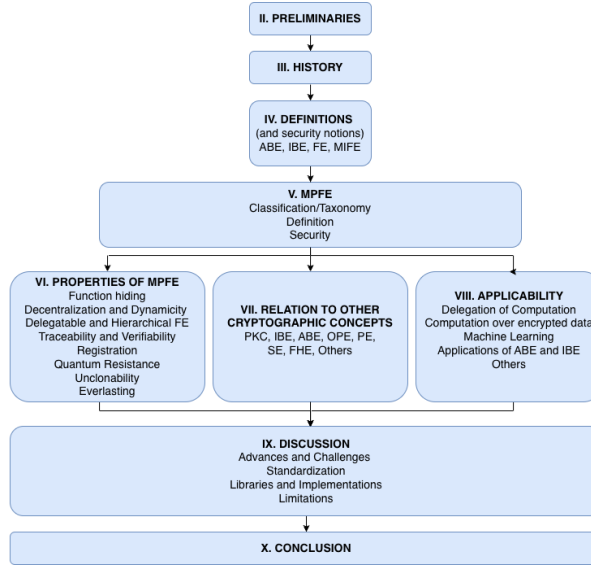


Figure 1: Paper flow

applicability of MPFE in real life. Section 9 discusses challenges, impact, standardization, and the current implementation status of MPFE schemes. Finally, the last section concludes. Figure 1 illustrates the flow of the paper.

2 Preliminaries

2.1 Acronyms and notations

Let $n > 1$ be a natural number. We consider $\{P_1, \dots, P_n\}$ a set of n parties (sometimes referred to as users), and $\mathcal{A}$ a PPT adversary. Let λ be the security parameter, and let f be a function (1-ary or n -ary, by case), with $f \in \mathcal{F}$, where $\mathcal{F}$ is a family of functions. We denote by $|x|$ the length of x , in bits. By $\forall$ and $\exists$ we understand the traditional notations *for all* and *exists*, respectively. Table 1 lists the acronyms used throughout the paper.

2.2 Basic definitions

Commonly, in cryptography, the adversaries are *Probabilistic and Polynomial in Time (PPT)*. This means that the adversary is a party (or a coalition of parties) that can run any algorithm that is polynomial in time and can use probabilistic choices, for example it can guess something (e.g., the plaintext, the key) or toss a coin (with uniformly random probability) to make a choice (e.g., choose from two possible plaintexts).

Usually, we call a scheme secure if any PPT adversary (regardless of how

Table 1: Acronyms

Acronym	Explanation
ABAC	Attribute-Based Access Control
ABE	Attribute-Based Encryption
aMIFE	Ad Hoc MIFE
CP-ABE	Ciphertext-Policy ABE
DDFE	Dynamic Decentralized Functional Encryption
QFE	Quantum Functional Encryption
DMCFE	Decentralized Multi-Client Functional Encryption
ECC	Elliptic Curve Cryptography
ETSI	European Telecommunications Standards Institute
FE	Functional Encryption
(F)HE	(Fully) Homomorphic Encryption
FL	Federated Learning
FPGA	Field-Programmable Gate Array
GPU	Graphical Processing Unit
HABE	Hierarchical Attribute-Based Encryption
HFE	Hierarchical Functional Encryption
HIBE	Hierarchical Identity-Based Encryption
IACR	International Association for Cryptologic Research
IBE	Identity-Based Encryption
IoT	Internet of Things
IP	Inner Product
IPFE	Inner Product Functional Encryption
KP-ABE	Key-Policy Attribute-Based Encryption
LWE	Learning with Errors
MA-ABE	Multi-Authority Attribute-Based Encryption
MAFE	Multi-Authority Functional Encryption
MCFE	Multi-Client Functional Encryption
MIFE	Multi-Input Functional Encryption
ML	Machine Learning
MLaaS	Machine Learning as a Service
MPFE	Multi-Party Functional Encryption
MQTT	Message Queue Telemetry Transport
NIST	National Institute of Standards and Technology
PE	Predicate Encryption
PKC	Public Key Cryptography
PKE	Public Key Encryption
PPFL	Privacy-Preserving Federated Learning
PPT	Probabilistic Polynomial Time
PSI	Private Set Intersection
QFE	Quantum Functional Encryption
SE	Searchable Encryption
SC	Secure Computation
SMPC	Secure Multi-Party Computation
(F)SS	(Functional) Secret Sharing
SXDH	Symmetric External Diffie-Hellman
TLS	Transport Layer Security
TTP	Trusted Third Party
U(F)E	Unclonable (Functional) Encryption
WLAN	Wireless Local Area Network

we instantiate it) has a *negligible* probability of winning a security game. In computational cryptography, negligible means too small for a PPT adversary to have real chances of winning. Usually, we refer to a negligible function in a security parameter as the inverse of an exponential function in the security parameter. The presence of a security parameter allows one to parameterize the probability of winning in real life, maintaining the security level in line with technological advances (the higher the security parameter, the lower the function's value). Although not always the case, the security parameter could be thought of as the length of the key.

Definition 2.1 (Negligible function [9]) *A function ϵ is called negligible (in the security parameter λ) if for all constants d there exists an λ_d such that for all $\lambda > \lambda_d$ it holds that $\epsilon(\lambda) < 1/\lambda^d$.*

PKE provides data confidentiality without requiring a shared secret key among the communicating parties. Let us suppose that one party P_1 wants to send a confidential message m to P_2 , and that P_2 has previously generated a public-private key pair (pk_2, sk_2) . Then, P_1 obtains pk_2 , the public key of P_2 (e.g., in a way that guarantees the authenticity of the key) and encrypts m using pk_2 to obtain $c = Enc(pk_2, m)$. When P_2 receives the ciphertext c , P_2 uses the private key sk_2 to decrypt $m = Dec(sk_2, c)$. Since P_2 is the only one who knows sk_2 , it is the only party that can recover the plaintext.

Definition 2.2 (Public-Key Encryption (PKE)) *A Public Key Encryption (PKE) scheme consists of three algorithms ($KeyGen, Enc, Dec$) defined as follows:*

- *KeyGen:* is a PPT algorithm that, when given a security parameter λ , outputs a secret key sk and a public key pk .
- *Enc:* is a PPT algorithm that, when given a plaintext m and a public key pk , outputs the ciphertext $c = Enc(pk, m)$.
- *Dec :* is a deterministic algorithm that, when given the ciphertext $c = Enc(pk, m)$ and the secret key sk , outputs $m = Dec(sk, c)$.

3 History

Diffie and Hellman introduced the concept of public-key cryptography back in 1976 [1,10]. The first public-key cryptosystem, currently one of the most widely used worldwide, RSA, appeared in the public domain a few years after [11]. Shamir defined IBE back in 1984 [12]. IBE is the first form in which FE appeared in the scientific literature. Since then, it has suffered several improvements in a series of works that define and further shape the concept of FE, starting with

the definition of ABE in 2005, by Waters and Sahai [13]. In 2020, the IACR awarded its Test-of-Time award to the paper [13] *"for laying the foundations of attribute-based encryption and other advanced notions of encryption"* [14]. This is not the single test-of-time award granted to a paper on FE. Previously, in 2016, another paper on attribute encryption [15], dated 2006 and authored by Goyal et al., won the CCS Test-of-Time award [16]. The first implementations of IBE appeared only many years after the introduction of ABE, in 2021 [17, 18]. Functional Encryption (FE), as a term, was first used in a written article in 2010 [19], although Sahai and Waters had previously used the term in their talks two years prior [20]. The indistinguishability-based security definition (IND-based security) was historically the first security notion considered for FE in 2005 [13]. The first framework for FE, including security definitions in both IND-based and simulation-based (SIM-based) settings, was given in [21].

Multi-Input Functional Encryption (MIFE) was independently introduced by Goldwasser et al. in [22] and Gordon et al. in [23] in the same year, 2013. The two papers were merged into a single one, published as [3]. The papers also tackle Multi-Client Functional Encryption (MCFE), a notion similar to MIFE, with the restriction that only ciphertexts labeled the same way (or encrypted with the same key as in [3, 23]) can be combined as input [24]. Multi-Authority Functional Encryption (MAFE) was defined shortly after, in [25] - as a generalization of the already existing Multi-Authority ABE (MA-ABE) [26], achieving FE in case of multiple authorities: each authority can certify a function (or credential) of its choice for a given user, and an user can find a function on the data if the trust of the data owner in the authorities providing the keys is "sufficient". Steps forward in comprising both decentralization and dynamicity have been made in the following years, e.g., Chotard et al. introduced Dynamic Decentralized FE (DDFE) [27] in 2020. In 2021, MPFE unified various flavors of FE that support multi-party, in the sense of distributed ciphertexts or secret keys, including MIFE, MCFE, and multi-authority schemes [4]. In the last few years, a significant number of papers on MPFE have been published. Most of these papers enhance or polish the existing results, such as refining definitions, improving efficiency, or expanding the pool of possible functions f . We refer to some of these papers in the following sections.

4 Definitions

We define Identity-Based Encryption (IBE) and Attribute-Based Encryption (ABE), Functional Encryption (FE) and Multi-Input Functional Encryption (MIFE), the first notion that historically generalized FE to multi-party settings.

4.1 Identity-Based Encryption (IBE) and Attribute-Based Encryption (ABE)

Identity-Based Encryption (IBE) is an encryption system for which any string can serve as a public key [2]. Typically, the public key is an IP address, an

email address, or a unique identifier that uniquely identifies the party, hence the name *identity-based*. For example, in an email system, the identity can be the email address, and the message can be the email content [5]. Similarly to FE, the system assumes the existence of a trusted authority that owns a master secret key msk . The authority receives the public key pk and, using the master secret key msk , generates a corresponding private key sk that fully decrypts any plaintext encrypted with pk [21].

Attribute-Based Encryption (ABE) generalizes IBE in that it allows a wider characterization of the party (or parties) intended to perform decryption, in terms of a set of attributes. It comes in two options [15]: *Ciphertext-Policy* (CP-ABE) and *Key-Policy* (KP-ABE) [2]. As stated in [6], the two flavors informally address the following questions: “*Who can access the data that I am encrypting?*” and “*As a decryptor, what type of data can I access?*”, respectively.

In the CP-ABE, the encryptor specifies a policy φ on the decryptor’s attributes, e.g., anyone who “is a student” and “is in the second year” and “studies engineering” is allowed to decrypt. Or, in a company, only someone with “employed_by > 10 years” and “location = HQ” is permitted to access the content of a special notification letter for a closed-door event. In terms of ABE encryption, this is denoted by $Enc(pk, (\varphi, m))$ for the public key pk , the policy φ , and any plaintext m . Contrary to CP-ABE, which places the access policy in the ciphertext, KP-ABE places the access policy φ in the key: the secret key $sk[f_\varphi]$ decrypts all encryptions $Enc(pk, (\mathbf{u}, m))$ for which $\varphi(\mathbf{u}) = 1$ [2]. By $\mathbf{u}$ here we understand the vector of attributes, as explained later.

Both IBE and ABE can be seen as specific cases of FE, although their scope is by definition different. We discuss more about the relation of IBE and ABE to FE in Section 7.

4.2 Functional Encryption (FE)

In FE, we assume an authority that holds a master secret key msk . For a function f from a family $\mathcal{F}$, the authority can generate an associated decryption key $sk[f]$, using the master key msk . Given an encryption of m , a party that knows $sk[f]$ can compute $f(m)$. More formally, let $c = Enc(pk, m)$ be an encryption of the plaintext m under the public key pk . Then, given the ciphertext c and the corresponding decryption key $sk[f]$ for a function f , the decryption outputs $Dec(sk[f], c) = f(m)$ [2].

Definition 4.1 (Functional Encryption (FE) [2]) *A FE scheme consists of four algorithms ($Setup, KeyGen, Enc, Dec$) defined as follows:*

- *Setup*: is a PPT algorithm that, when given a security parameter λ , outputs a master secret key msk and a public key pk .
- *KeyGen*: is a PPT algorithm that, when given a security parameter λ , a master secret key msk , and a function f , outputs an associated secret key $sk[f]$.

- *Enc*: is a PPT algorithm that, when given a plaintext m and the public key pk , outputs the ciphertext $c = \text{Enc}(pk, m)$.
- *Dec*: is a deterministic algorithm that, when given the ciphertext $c = \text{Enc}(pk, m)$ of a plaintext m and the decryption key $sk[f]$, outputs $f(m) = \text{Dec}(sk[f], c)$.

4.3 Security of FE

The security of FE is usually defined in computational settings, i.e., against a PPT adversary as described in Section 2. Intuitively, the FE is secure if the holder of the secret-key $sk[f]$ can learn nothing else about m except $f(m)$ (up to a negligible probability). The security should also stand against coalitions of parties: if an adversary gains access to a set of decryption keys $sk[f_1], \dots, sk[f_t]$, the adversary should learn nothing else except $f_1(m), \dots, f_t(m)$ [2, 5].

There are two main models of security for an FE scheme: (1) indistinguishability-based security, for short IND-based security, and (2) simulation-based security, for short SIM-based security [5]. SIM-based security is stronger than IND-based security, in the sense that any scheme proved secure in the SIM-based security model is also secure in the IND-based security model. The reverse does not hold; there are IND-secure FE schemes that are not SIM-based secure [5].

In the IND-based security model, no PPT adversary can distinguish between ciphertexts of any two messages m_0 and m_1 of its choosing and of the same length, having access to some secret keys $sk[f_i]$ for functions f_i such that $f_i(m_0) = f_i(m_1)$. The adversary is restricted from obtaining $sk[f]$ to prevent trivial winning. The security notion is a straightforward generalization of the IND-based definition of PKE, expressed as a game between the adversary $\mathcal{A}$ - who tries to break the system, and a challenger $\mathcal{C}$ - who challenges the adversary to break the system with the aim to prove its strength. We further give a strong security definition that allows the adversary to be *adaptive* in the sense that it can choose the challenges based on the received answers.

Definition 4.2 (FE IND-based security [5]) *Let $\mathcal{A}$ be a PPT adversary that receives the public key pk and can adaptively request a polynomial number of keys $sk[f_i]$, each corresponding to a function f_i . The adversary $\mathcal{A}$ then sends to the challenger $\mathcal{C}$ two distinct messages of the same length m_0 and m_1 such that $f_i(m_0) = f_i(m_1)$, for all i . The challenger chooses a bit b uniformly at random, encrypts $c = \text{Enc}(pk, m_b)$, and returns c to $\mathcal{A}$. The adversary $\mathcal{A}$ continues to query $\mathcal{C}$ for $sk[f_i]$, satisfying the same property as above, i.e., $f_i(m_0) = f_i(m_1)$. Finally, $\mathcal{A}$ outputs a bit b' and wins if it guesses $b' = b$. The scheme is IND-based secure if the winning probability is upper bounded by $1/2 + \epsilon(\lambda)$, with $\epsilon(\lambda)$ negligible in the security parameter λ .*

The SIM-based security definition asks that the knowledge of $\mathcal{A}$ on m is the same as the knowledge of $\mathcal{S}$ on m , where $\mathcal{A}$ knows the ciphertext c and the keys $sk[f_1], \dots, sk[f_t]$, while $\mathcal{S}$ only knows the values $f_1(m), \dots, f_t(m)$. This means

that the adversary $\mathcal{A}$ cannot learn anything else from the ciphertext and the set of keys, except the corresponding functional decryptions, which $\mathcal{A}$ is allowed to know by definition [2].

Definition 4.3 (FE SIM-based security [21]) *Let $\mathcal{A}$ be a PPT adversary that takes as input the public key pk , a set of keys $sk[f_1], \dots, sk[f_t]$ corresponding to functions $f_1, \dots, f_t$ of its choice, and a ciphertext $c = \text{Enc}(pk, m)$ for a plaintext m . The FE system is SIM-based secure if for all PPT adversaries $\mathcal{A}$ defined as above, there exists another polynomial algorithm $\mathcal{S}$, called the simulator, which, given pk and $f_1(m), \dots, f_t(m)$ (but not c), can output the exact information about m as $\mathcal{A}$.*

Something that FE leaks, and in fact every cryptosystem leaks, is the length of the plaintext given the ciphertext. Although techniques to hide (to some extent) the length can be applied (for example, an extended padding), they raise efficiency problems and thus are useless in practice. As in [2], we refer to the information leaked by c on m as the *empty functionality* and denote it by f_ϵ . Thus, $f_\epsilon(m) = |m|$ signifies that the ciphertext leaks the length of the plaintext [21]. The privacy of the function - meaning that the function itself should not leak information, thus, that $sk[f]$ does not reveal information about the functionality of f - was first considered in the settings of private-key predicate encryption [28] and later adopted for FE [5].

A more in-depth definition of FE and its security is available in [21, 29]. Older papers, including [21] and [30], have discussed limitations with respect to the SIM-based security of FE. The security of both FE and MIFE is addressed in [31].

4.4 Multi-Input Functional Encryption (MIFE)

MIFE is a generalization of FE for which the function f is n -ary, so it takes as input n plaintexts, $n > 1$. Typically, n is polynomial in the security parameter λ . Similarly to FE, we denote by $sk[f]$ a secret key that corresponds to the function f , thus $sk[f]$ allows one to compute $f(m_1, \dots, m_n)$ given $(c_1, \dots, c_n)$, with $c_i = \text{Enc}(pk_i, m_i)$, for all $i = 1, \dots, n$, with respect to the same master secret key msk [22]. The public keys used to encrypt the n plaintexts can be equal $pk_1 = \dots = pk_n$ (*symmetric key settings*) or distinct (*asymmetric key settings*). The latter case captures settings in which each encryption is performed by a different party [22].

Definition 4.4 (MIFE [3]) *A Multi-Input Functional Encryption (MIFE) scheme consists of four algorithms ($\text{Setup}, \text{KeyGen}, \text{Enc}, \text{Dec}$) defined as follows:*

- *Setup:* is a PPT algorithm that, when given a security parameter λ and the function arity n , outputs a master secret key msk , a set of n public keys $pk_1, \dots, pk_n$.

- *KeyGen*: is a PPT algorithm that, when given a security parameter λ , a master secret key msk , and an n -ary function f , outputs an associated secret key $sk[f]$.
- *Enc*: is a PPT algorithm that, when given a plaintext m and a public key pk_i in the set of public keys $\{pk_1, \dots, pk_n\}$, outputs the ciphertext $c = Enc(pk_i, m)$.
- *Dec*: is a deterministic algorithm that, on given the n ciphertexts $c_i = Enc(pk_i, m_i)$ of plaintexts m_i , $i = 1, \dots, n$ and the decryption key $sk[f]$, outputs $f(m_1, \dots, m_n) = Dec(sk[f], c_1, \dots, c_n)$.

In fact, in [3], the correctness of the above scheme (i.e., the decryption of $(c_1, \dots, c_n)$ using the key $sk[f]$) holds only up to a negligible probability (in the security parameter λ). Our definition above is stricter, in the sense that the probability of an error at decryption is zero. We view this as a technicality that does not affect the understanding of the current paper but instead makes it easier to comprehend. As given in [3], the definition includes the case in which all public keys are equal, i.e., $pk_1 = \dots, pk_n$. In this case, an additional index is required to maintain the given order of the ciphertexts. Again, this is a technicality omitted in both this paper and [3], without impact on our current presentation.

4.5 Security of MIFE

Same as for the security of FE, we refer to two distinct definitions of security: IND-based security and SIM-based security [3, 22]. In fact, both definitions are generalizations of the security definitions of FE, as described in Section 4.3.

In IND security, we consider a game between a challenger $\mathcal{C}$ and an adversary $\mathcal{A}$. The challenger generates the master secret key msk and the encryption keys $pk_1, \dots, pk_n$, then gives to the adversary a subset of encryption keys, chosen by the adversary. The adversary can then request any number of secret keys $sk[f]$ for functions f of the adversary's choice. Next, the adversary declares two challenges: two vectors $\mathbf{m}^0$ and $\mathbf{m}^1$, where $\mathbf{m}^i$, $i = 1, 2$ is a set of plaintexts. The challenger randomly chooses a bit $b \in \{0, 1\}$, and encrypts each element of $\mathbf{m}^b$ using the encryption key pk_i at position i to obtain a tuple of ciphertexts $\mathbf{c}^b$, which is then given to the adversary. Again, the adversary can request any polynomial number of secret keys $sk[f]$ for functions f of the adversary's choice. During the whole game, the adversary is restricted to ask for secret keys such that there exist vectors of plaintexts $\hat{\mathbf{m}}^0$ and $\hat{\mathbf{m}}^1$ where for every $i = 1, \dots, n$, either $\hat{m}_i^b \in \mathbf{m}_i^b$ or the adversary has pk_i , such that $f(\hat{\mathbf{m}}^0) \neq f(\hat{\mathbf{m}}^1)$. This prevents trivial winning. Finally, the adversary has to guess the bit b that the challenger chose. The adversary wins if the probability to win (i.e., guess $b' = b$) is negligibly larger than $1/2$ [3].

In SIM security, there are two natural settings to consider: (1) the adversary has access to the encryption key(s), analogous to the public-key settings, and (2) the adversary does not have access to any encryption key, analogous to the

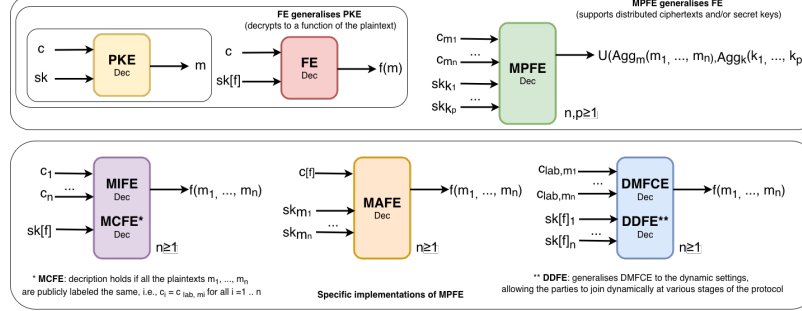


Figure 2: Graphical illustration of decryption in MPFE and related notions (inspired by [32])

secret-key settings [3]. Of course, security against adversaries, as defined in the first setting where the adversary has access to the public keys, is stronger. Regardless of the case, in SIM-based security, the adversary can be simulated using only oracle access to the functions f for which the adversary obtains secret keys $sk[f]$. This holds even if the adversary can obtain a set of challenge ciphertexts corresponding to unknown plaintexts.

The security of MIFE in SIM security settings is challenging and full of negative results. In particular, the feasibility of constructing a simulation-based secure MIFE reduces to the existence of virtual black-box obfuscation, which is known to be impossible [3]. The reader can refer to [3, 5, 22, 23, 31] for details.

5 Multi-Party Functional Encryption (MPFE)

Based on [4], we consider the broader notion of MPFE, which encompasses a range of schemes that handle distributed inputs (ciphertexts) and keys. To improve readability, we first discuss how MPFE unifies related concepts before giving its definition and discussing its security.

5.1 Classification

Figure 2 illustrates the differences in MPFE regarding the decentralization of ciphertexts, keys, or both, and following the line of [4], in relation to related notions. Since the functionality lies in the decryption, inspired by [32], we only depict the decryption operation. Here, U is a universal function, and Agg_m and Agg_k are scheme-specific functions that aggregate the ciphertexts and keys respectively.

The motivation for multiple inputs (ciphertexts) is multifaceted, allowing, for example, sensitive data originating from multiple sources, each encrypted under a distinct public key. Multiple use cases for the n -ary input and its distribution among multiple parties are given in Section 8. The motivation for the decentralization of keys is mainly dual: (1) key escrow - to avoid a single

party that is in full control of the master key, and thus becoming a single point of attack against which no security can hold and (2) better fitting in the real world - in which there usually are multiple authorities, each responsible for either issuing different attributes or each corresponding to different organizations, companies, countries, etc. that do not necessarily trust each other [4].

To accommodate the setup in all its forms, MPFE accepts three modes [4]: (1) *central* - a TTP outputs the master secret keys and the encryption keys for all the parties in the system, (2) *local* - each party runs the setup independently and outputs its own encryption key and/or master secret key, and (3) *interactive* - the setup is an interactive protocol run by a set of parties, at the end of which, each party has its encryption key and/or master secret key.

Table 2 provides brief definitions of the main MPFE notions, includes an informal description highlighting their key properties, and references the papers that introduced them. Figure 3 provides a unified taxonomy diagram that visually organizes the discussed concepts.

5.1.1 Distributed ciphertexts

MIFE is an example of an MPFE with distributed ciphertexts, as it supports multiple inputs. Recall that MIFE allows different parties $P_1, \dots, P_n$ with inputs $m_1, \dots, m_n$ to compute a joint function $f(m_1, \dots, m_n)$ in the following way: each party P_i encrypts m_i to obtain c_i , then the decryptor who owns a functional key $sk[f]$ (derived based on the master key msk by an authority) can compute $f(m_1, \dots, m_n)$ given $c_1, \dots, c_n$ [3, 4].

Multi-Client Functional Encryption (MCFE) is defined similarly, except that the inputs m_i are additionally associated with public labels lab_i , and they can be combined only with other inputs with the same label [3, 4].

5.1.2 Distributed keys

This setting corresponds to multiple authorities, which normally run their own setup to generate public-private key pairs. In the case of ABE, the keys can either correspond to the same attribute att , so the keys associated with the same attribute are generated by independent authorities, or each authority can be responsible for different sets of attributes [4].

The first case denotes the Decentralized Attribute-Based Encryption with Policy Hiding (DABE) [33], in which an encryptor encrypts a message m along with a general access structure C , and independent authorities issue the secret keys corresponding to (the same) attribute att . Decryption returns the message m if $C(att) = 1$, and the access policy in the ciphertext is hidden [4].

The second case is illustrated by the Multi-Authority Functional Encryption (MAFE) [25], a generalization of Multi-Authority ABE (MA-ABE) [26, 34]. Consider an encryptor that computes a ciphertext for a message m along with a policy f over the various authorities. Any authority A_i , $i = 1, \dots, n$ can generate a token for a user for attributes lab_i . A decryptor with tokens for lab_i from authority A_i , can decrypt the ciphertext to recover $f(lab_1, \dots, lab_n, m)$ [4]. The

Table 2: **Main MPFE-related concepts**

Notion	Distr. ci-pher-texts	Distr. keys	Authority	Description and Ref.
MPFE	Yes	Yes	Yes (Central Setup) No (Local and In-teractive Setup)	Multi-Party Functional Encryption: Unifies and abstracts various notions of FE that support distributed ciphertexts and/or secret keys, including the notions that follow in this table. [4]
PKE	No	No	No	Public Key Encryption (PKE): Performs encryption and decryption asymmetrically, using a pair of public-private keys. [1]
FE	No	No	Yes (central)	Functional Encryption (FE): generalizes PKE by decrypting to a function of the plaintext $f(m)$, rather than simply exposing the plaintext m . [19]
MIFE	Yes	No	Yes (central)	Multi-Input Functional Encryption: Allows the computation of $f(m_1, \dots, m_n)$ by a decryptor that owns a decryption key $sk[f]$ (derived from a master secret key msk) and receives as input $c_1, \dots, c_n$, where c_i is the encryption of m_i by the party P_i , $i = 1, \dots, n$. [3]
MCFE	Yes	No	Yes (central)	Multi-Client Functional Encryption: Similar to MIFE, but the inputs are (publicly, typically time-based) labeled and can only be combined if they are labeled the same. [3]
MAFE	No	Yes	Yes (multi-authority)	Multi-Authority Functional Encryption: Generalizes both FE and MA-ABE, considering multiple authorities, where each authority can certify a function (or credential) of its choice for a given user, and a user can find a function of the data if the trust of the data owner in the authorities providing the keys is sufficient. [25]
DMCFE	Yes	Yes	No	Decentralized Multi-Client Functional Encryption: Removes the central authority in MCFE, making the clients to work together to generate appropriate functional decryption keys, thus the setup phase and the generation of functional decryption keys are decentralized, among the same clients as the ones that generate the ciphertexts. [24]
DDFE	Yes	Yes	No	Dynamic Decentralized Functional Encryption: Generalises DMCFE by allowing the parties to join at various stages during the lifetime of a system without the need of a TTP, while maintaining input decentralization from multiple users. [27]

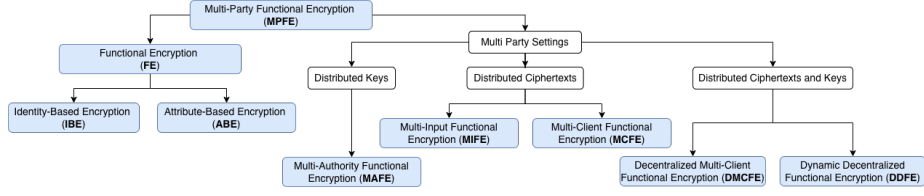


Figure 3: Unified taxonomy diagram

main difference between MAFE and MA-ABE [34] is the same difference between FE and ABE, in (MA-)ABE, the result is in plaintext, while in (MA)FE the result is a function on the plaintext [25].

5.1.3 Distributed ciphertexts and keys

Finally, the distribution of both ciphertexts and keys can be combined in several ways. Such a solution supports both distributed data sources (i.e., the input ciphertexts originate from different parties) and multiple authorities to deal with the keys. Examples of such schemes include [4]: Decentralized Multi-Client Functional Encryption (DMCFE) [24], Ad Hoc MIFE (aMIFE) [35], and Dynamic Decentralized FE (DDFE) [27]. In particular, DMCFE [24] examines the case where there is no single master secret key, but the parties establish their individual (correlated) master secret keys as a result of a setup protocol. DDFE [27] generalizes DMCFE by allowing parties to join the system dynamically, without relying on a TTP or on expensive MPC. We will discuss decentralization and dynamicity further in the following sections.

Regarding the functionality of f , MPFE schemes (in different settings, security models, or particular assumptions) can accommodate different functions, including the Inner Product (IP), quadratic functions, or arbitrary circuits. More details are available in [5, 36].

Table 3 gives a brief comparison of the main FE-related concepts (in terms of the hardness assumption and the setup overhead). Note that the comparison is based on the initial references, also indicated in Table 2, and focuses on the IP functionality, for which the implementations are practical.

5.2 Definition on MPFE

Let n be the number of ciphertext inputs and p the number of key inputs. Let $\mathcal{C} = \mathcal{C}_{pub} \times \mathcal{C}_{priv}$ be the space of ciphertext inputs and $\mathcal{K} = \mathcal{K}_{pub} \times \mathcal{K}_{priv}$ be the space of key inputs. Note that the ciphertexts and the keys have a public (*pub*) and a private (*priv*) component each. One could see the public component as the index and the private component as the corresponding private value. More precisely, the input message could be thought of as $m_i = (m_{pub,i}, m_{priv,i}) = (i, m^i)$, where m^i is the actual plaintext of the party i , $1 \leq i \leq n$. Similarly, $k_j = (k_{pub,j}, k_{priv,j}) = (j, k^j)$, where k^j is the private key of the authority j ,

Table 3: **Comparison of the main FE-related concepts.** (n denotes the number of parties and a is the number of authorities in MAFE)

Ref.	Notion	Hardness Assumption	Setup	Setup Cost
[37]	FE	DDH, LWE	Central Setup/ Centralized	$\mathcal{O}(1)$
[3]	MIFE	DDH	Central Setup/ Centralized	$\mathcal{O}(n)$
[3]	MCFE	DDH	Central Setup/ Centralized	$\mathcal{O}(n)$
[25, 34]	MAFE	SXDH	Central Setup/ Multi-Authority	$\mathcal{O}(a)$
[24]	DMCFE	DDH	Interactive Setup	$\mathcal{O}(n^2)$
[27]	DDFE	DDH, LWE	Local Setup/ Dy- namic	$\mathcal{O}(\log(n))$

$1 \leq j \leq p$. Moreover, the primitive considers encryption keys for n encryptors and master keys for p authorities, thus embodying the distributed character in both ciphertexts and keys. Two aggregation functions $Agg_{ct} : \mathcal{C}^n \rightarrow \mathcal{C}$ and $Agg_k : \mathcal{K}^p \rightarrow \mathcal{K}$, defined for the ciphertext secret-key input spaces, respectively, specify how the inputs can be combined to capture a specific primitive. Thus, the aggregation functions are scheme-specific. This captures both ciphertext and key policy schemes, as either m_i and k_i can be descriptions of functions [4].

The *Setup* captures the centralized, local, and interactive (or decentralized) setups, as discussed in Section 5.1 and illustrated in Figure 4. The *KeyGen* and the *Enc* algorithms are run independently by each of the p authorities and n users, respectively, while *Dec* is an algorithm run only once. Note that if the setup is interactive, then the *KeyGen* and the *Enc* can be non-interactive.

Definition 5.1 (MPFE [4]) *A Multi-Party Functional Encryption (MPFE) scheme consists of four algorithms (*Setup*, *KeyGen*, *Enc*, *Dec*) defined as follows:*

- *Setup:* is a PPT algorithm that, when given a security parameter λ , the number of ciphertexts inputs n , the number of key inputs p , and two aggregation functions Agg_{ct} and Agg_k , outputs a set of master secret keys $\{msk_1, \dots, msk_p\}$, a public key pk , and a set of encryption keys $\{pk_1, \dots, pk_n\}$.

It can run in three modes {Central, Local, Interactive}:

- *Central:* The *Setup* is run by one TTP which outputs the master secret keys and the encryption keys for all the parties in the system.
- *Local:* The *Setup* is run independently by different parties, without

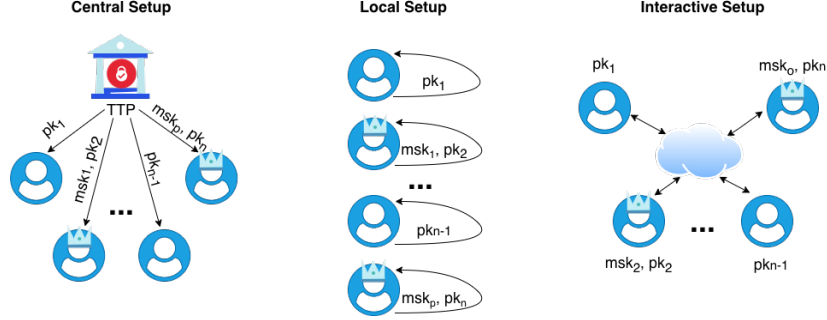


Figure 4: Graphical illustration of the three types of setup in MPFE. For simplicity of illustration, the figure considers n basic users who get an encryption key pk and p authorities that get both an encryption key pk and a master secret key msk .

interactions, and each party outputs its own encryption key and/or master secret key.

– *Interactive: The Setup is an interactive protocol run by a set of parties so that at its termination each party has its own encryption key and/or master secret key.*

- *KeyGen: is a PPT algorithm run independently by all p key authorities that, when given a public key pk , a master secret key msk_j , the user index j , $1 \leq j \leq p$, and a function input $k_j = (k_{pub,j}, k_{priv,j})$, it outputs a secret key sk_j .*
- *Enc: is a PPT algorithm that is run interdependently by the n users that, when given the public key pk , an encryption key pk_i , the user index i , $1 \leq i \leq n$ and an input $m_i = (m_{pub,i}, m_{priv,i})$, it outputs a ciphertext c_i .*
- *Dec: is a deterministic algorithm that, given the partial keys $\{sk_i\}_{i \leq p}$ and partial ciphertexts $\{c_j\}_{j \leq n}$ outputs $\mathcal{U}(\text{Agg}_{ct}(\{m_i\}), \text{Agg}_k(\{sk_i\}))$, where $\mathcal{U}$ is the universal circuit, or $\perp$.*

A special case is the dynamic notion of MPFE. A fully *dynamic* MPFE allows the number of inputs (keys and ciphertexts) to be unspecified during setup, and for the aggregation functions to be specified only during key generation and encryption. Because the number of parties is unknown at setup, in dynamic MPFE we are interested in the local setup, where the parties run it independently and output their own keys [4].

5.3 Security of MPFE

The indistinguishability security definition of MPFE is similar to the one of MIFE but takes into account corruption queries [4]. This is natural, as cryptographic keys are decentralized and thus corruption of some of them has to be

permitted. More precisely, in MPFE the adversary has access to three oracles: (1) *Corrupt oracle*, which allows the adversary to ask for the encryption key pk or the master secret key msk of any party or authority, (2) *Key oracle*, which samples decryption keys for parties and functions on the adversary’s choice, if the party has not been already corrupted, and (3) *Encryption oracle*, which samples ciphertexts for parties that have not been previously corrupted. The key and encryption oracles are similar to MIFE, while the corrupt oracle is new to MPFE. Note that in case of local setup, the adversary can simply supply the oracle with adversarial generated keys for the corrupted users, while for the interactive setup the adversary can simulate the behavior of the corrupted party in the setup protocol. Normal restrictions against the corrupt oracle are further enforced in the security experiment such that the adversary cannot trivially win (e.g., corrupt all necessary parties such that it can trivially respond to the challenge) [4].

Similarly, in dynamic settings, the oracles in the security game are defined accordingly, and include an additional oracle that generates a fresh tuple of key pairs and a master key that are stored in a list. This models the dynamic join in the system at any stage. Again, the dynamic settings only consider the local setup mode [4].

6 Properties of MPFE

We informally discuss some properties MPFE schemes can possess, and which we find valuable for the applicability of such schemes in the real world. Table 4 summarizes the properties.

6.1 Function hiding

Firstly, as already mentioned in Section 4.3, the traditional definition of FE does not necessarily hide the function f itself. However, the function can contain or expose sensitive data. Function hiding has been considered in both single-user and multi-user settings and for different function types (e.g., inner products, quadratic functions) [38, 39]. Recent papers recognize function hiding as an important requirement [4, 40–42]. For example, function hiding enables secure delegation of computation to an untrusted server without the server learning the functionality. This is important because, in many cases, knowing the functionality and the output $f(m_1, \dots, m_n)$ might leak information about the underlying data $m_1, \dots, m_n$. Also, there might be the case that the functionality should remain private because of copyright [4].

Commonly, FE can be defined in two flavors: *public-index* and *private-index* (or *non-public index*), and the message to be encrypted is of the form (ind, m) , where ind is an index (e.g., attributes or policies in ABE). In public-index FE, the index is public, while in private-index FE, it is not. Thus, the empty functionality is defined by $f_\epsilon(ind, m) = (ind, |m|)$ in public-index FE and by

Table 4: **Properties**

Name	Description
Function hiding	Hides the functionality/definition of f .
Decentralization	Allows the decentralization of the input to multiple parties (users) and/or the decentralization of keys (for example, by replacing a single, trusted authority with multiple authorities or, to some extent, the users themselves).
Dynamicity	Allows parties (users, authorities, etc.) to join (and preferably also leave) the system at various stages (ideally, all possible states).
Revocation	Supports the revocation of private keys (or, more generally, rights of parties, authorities, etc.)
Delegatability	Permits users to derive decryption keys for policies that are more restrictive than their own; generalizes Hierarchical FE (HFE).
Traceability	Allows for the identification of the malicious party/colluding parties.
Verifiability	Ensures that even if the ciphertext is maliciously generated, and the setup and the key generation are malicious, the correctness of the decryption is meaningful.
Registration	Allows a party to sample its own public-private key pair, register the public key and a function f with the key curator, and later decrypt an encryption of m to learn $f(m)$ using the secret key and a helper secret key from the key curator.
Quantum resistance	Ensures that the security is maintained in the presence of quantum adversaries too.
Unclonability	Guarantees that two parties cannot simultaneously recover the correct function outputs using two independently sampled function secret keys.
Everlasting	Guarantees computational security during the run of the protocol (and some time after) and unbounded computational security after the protocol has terminated; it is meaningful in the quantum world.

$f_\epsilon(ind, m) = |m|$ in non-public FE. In the latter case, the aim is to reveal as little as possible [21].

6.2 Decentralization and Dynamicity

We have already discussed decentralization, both in terms of distributed ciphertexts and distributed keys. We emphasize, once more, the role of multiple authorities in eliminating a single point of trust (an authority cannot decrypt without the cooperation of other authorities) and in increasing the system’s scalability, so that failures or disruptions of some authorities should not affect others [5]. A valuable property is the ability to allow authorities to join the system without requiring interaction. In ABE, this was achieved in [34]. This ability already accounts for dynamicity, as authorities can participate dynamically in the protocol execution.

The definition of MPFE in [4] assumes, for simplicity of presentation and to remain aligned with most existing work, that the number of parties, as well as the aggregation functions, are input to a setup algorithm. However, full dynamic schemes can be integrated into the same definition, allowing parties to join the protocol without prior agreement and even choose the functionality dynamically so that it can change over different instances of the protocol [4].

Another aspect closely related to dynamicity is the ability to revoke private keys. Revocation has been considered in both IBE [43] and ABE [44] in an efficient and scalable manner.

Dynamicity and decentralization are often simultaneously required. Thus, Dynamic Decentralized Functional Encryption (DDFE), introduced in [27], naturally generalizes FE by incorporating both dynamicity and decentralization: multiple users join the system dynamically without interaction and without relying on a TTP or heavy MPC. DDFE is not the only concept that attempts to jointly accommodate decentralization and dynamicity (see Section 5.1, Distributed ciphertexts and keys). Other notions have been defined, including aMIFE [35] and Multi-Input Non-Interactive Functional Encryption (MINIFE) [45]. The concept of DDFE generalizes the Decentralized Multi-Client Functional Encryption (DMCFE) [24] in the sense of a non-interactive global setup that outputs some public parameters used by each party that joins the system to run a local setup algorithm to generate their own secret key [27, 40]. Very recently, the work [40] advances DDFE by defining generic compilers that improve upon the state-of-the-art in terms of security, computational assumptions, and functionality.

6.3 Delegatable and Hierarchical FE (HFE)

As a natural generalization of HIBE and HABE, Hierarchical Functional Encryption (HFE) was proposed. In fact, any FE can be transformed into an HFE using a generic transformation, which was introduced in [46]. In an HFE scheme, the holder of a key $sk[f]$ for a function f can generate a key $sk[g \circ f]$ corresponding to the function $g \circ f$ for any function g [5]. The party that holds

$g \circ f$ can compute $g(f(c))$ given a ciphertext c , but cannot find anything about the plaintext m or the value $f(m)$ (except, of course, what is leaked by the definition of g itself if the scheme is not function hiding).

HIBE encompasses the idea of delegatable functional encryption, also considered in [25]. In delegatable FE, any user can derive a decryption key for a policy that is more restrictive than its own [25].

6.4 Traceability and Verifiability

In conventional FE, possession of a decryption key $sk[f]$ is sufficient to perform decryption. Moreover, for a given function f , more parties authorized to decrypt typically possess the same key $sk[f]$. This raises a problem, in that it becomes impossible to identify a malicious party that exposes the key among the set of parties holding the key [5, 47]. Traceability makes this possible, leading to the identification of the malicious party or colluding parties. To achieve this, a simple mechanism is to bind the decryption key not only to the function f , but also to the party. Traceable functional encryption was introduced in [47].

Another possible act of a malicious party is to deviate from the correct execution of the protocol (without being detected). A scheme that overcomes this is called verifiable. Verifiability has been previously considered for other cryptographic primitives, such as secret sharing, which inspired its adoption in FE [48]. In FE, verifiability ensures that even if the ciphertext is maliciously generated and the setup and key generation are malicious, the correctness of decryption is still meaningful. This captures the case of a malicious authority and a malicious encryptor [48].

6.5 Registration

Registered Functional Encryption (RFE), introduced in [49, 50] and an extension of existing works (see Sections 7.2 and 7.3), eliminates the need for trust in the central authority to handle secrets. Unlike a standard FE, in an RFE scheme, the role of the trusted authority is greatly diminished. The parties create their own secret keys, register the associated public keys and the functions to a *key curator*, which aggregates them into a single master key. The key curator then occasionally helps parties with helper decryption keys that the parties combine with their own secret keys [49]. RFE was also constructed on lattices [51].

6.6 Quantum resistance

Based on their hardness assumptions, cryptographic schemes can be secure in classical settings (i.e., the assumptions hold against classical adversaries) or quantum-resistant (i.e., the assumptions hold against quantum adversaries, too). Mathematical problems that are strong candidates against quantum adversaries include lattice-based problems, such as Learning with Errors (LWE) or the Shortest Vector Problem (SVP), hash-based problems that rely on the

difficulty of inverting hash functions, and code-based problems originating in coding theory. We discuss quantum resistance further in Section 9.1.

6.7 Unclonability

The work [52] extends FE to the quantum setting. It provides a construction for a Quantum Functional Encryption (QFE) scheme, built on quantum garbled circuits and allowing for the evaluation of polynomial-sized circuits on arbitrary quantum messages. By construction, a QFE scheme universally satisfies Unclonable Functional Encryption (UFE). UFE guarantees that two parties cannot simultaneously recover the correct function outputs using two independently sampled function secret keys. UFE generalizes the concept of Unclonable Encryption (UE) [53] in quantum cryptography, which guarantees that an adversary in possession of a ciphertext cannot generate two states that correctly decrypt to the same message.

6.8 Everlasting

The paper [54] introduces certified everlasting FE. In everlasting cryptography, meaningful in the quantum world, the adversary must succeed during the protocol’s execution; otherwise, after the protocol has terminated, security holds unconditionally [55]. In the case of FE, a decoder that knows the ciphertext of a message m and the functional decryption key $sk[f]$ can obtain $f(m)$ and nothing else [54]. This holds for unbounded adversaries starting at some point after the protocol’s completion. For the execution of the protocol, the adversary remains computationally bounded [55].

7 Relation to other cryptographic concepts

We further clarify the relation among (MP)FE, and other cryptographic concepts. We first start with the observation that FE is a particular case of MIFE, for which $n = 1$, and thus, a particular case of MPFE. Additionally, IBE and ABE are specific implementations of FE, as previously mentioned. However, we recall these notions here and emphasize, again, the relation among them. Table 5 summarizes the main goals that the discussed cryptographic primitives aim to achieve.

7.1 Public key cryptography

A traditional public-key encryption system is a particular case of FE, for which the function f is the identity function: $f_{id}(m) = m$, for all possible plaintexts m . In this case, the private key in traditional public-key settings is simply $sk[f_{id}]$ [2].

Table 5: **Related cryptographic concepts**

Acronym	Description
(MP)FE	Allows a party that holds a secret key $sk[f]$ to learn $f(m)$, a function on the plaintext m , and (ideally) nothing else, from the corresponding ciphertext; it is not interactive once $sk[f]$ is known.
PKE	Allows a party that owns a private key sk to learn the plaintext m from the corresponding ciphertext, encrypted using the corresponding public key pk .
IBE	Generalizes PKE in the sense that the public key can be any string, usually the identity.
ABE	Generalizes IBE in the sense that it allows a more general characterization. of the party intended to perform decryption, in the form of a set of attributes.
OPE	Preserves the order of the plaintexts in the ciphertexts.
ORE	Generalizes OPE in the sense that it publicly (i.e., without decryption) reveals the order of the underlying plaintexts from the ciphertexts.
PE	Allows decryption to parties whose set of attributes satisfies a predicate.
SE	Enables search operations on encrypted data without decryption.
FHE	Allows computation on the encrypted data, in the sense that, given some ciphertexts, a party learns the encrypted result of a computation on the corresponding plaintexts.
(MP)SC	Allows one authorized party that owns a secret key to learn a function on the plaintext(s), being similar to (MP)FE but interactive.
PSI	Permits the computation of the intersection of sets belonging to different parties without revealing anything else except the intersection.
FSS	Generalizes secret sharing in the sense that the parties can compute $f(s)$ for a function f in a family of functions, instead of simply computing the secret s .

7.2 Identity Based Encryption (IBE)

IBE can be implemented using FE only; thus, it is a particular case of FE. To build an IBE from FE, let us assume msk the master secret key of the authority, id the public key in the IBE system, pk the public key in the FE system, and the function $f_{id'}(id', m) = m$, if $id = id'$ and $\perp$ otherwise. The key $sk[f_{id}]$ in the FE system allows the party holding the public key id only to decrypt ciphertexts corresponding to its identity, defined in the IBE and FE system as $Enc'(id, m) = Enc(pk, (id, m))$ [2].

A particular case of IBE is the Hierarchical IBE (HIBE), for which identities are vectors that model the hierarchical structure of an organization. The dimension of the identity vector indicates the level in the hierarchy where that identity is positioned, with the first level being the root. An identity on a given level l can derive private keys for its descendants [5]. IBE might have other particular properties, for example, hiding the ciphertext identity, as in anonymous IBE [5, 17].

Recently, significant advances have been made in IBE-related research, including big-key IBE [56] and Registration-Based Encryption (RBE) [57]. In Big-Key IBE, the idea is to defend against key-exfiltration attacks by making the cryptographic secrets large enough that it becomes infeasible for an adversary to exfiltrate the key without being detected. The work [56] proposes the first adaptively secure construction of a big-key IBE from standard cryptographic assumptions. RBE [58] is a primitive that attempts to combine the best aspects of PKE (self-generation of keys) and IBE (usage of the identity as the public key). RBE replaces the trusted authority in IBE with a key curator responsible for aggregating the public keys in a way that mimics IBE functionality, but whose role is limited to aggregation and storage, not the complete generation of keys.

More information on IBE and HIBE can be found in [5, 59, 60].

7.3 Attribute Based Encryption (ABE)

Again, ABE can be implemented using FE only; thus, it is a particular case of FE. To build a CP-ABE from FE, let us assume msk the master secret key of the authority, φ the attribute policy in the ABE system, pk the public key in the FE system, and the function $f_{\mathbf{u}}(\varphi, m) = m$, if $\varphi(\mathbf{u}) = 1$, and $\perp$ otherwise. By $\mathbf{u}$ we denote a binary vector to characterize the holding of attributes, i.e., for a set of attributes, the position is 1 if the corresponding attribute in the set holds, and 0 otherwise. By $\varphi(\mathbf{u}) = 1$, we understand that all required attributes in φ are satisfied. The key $sk[f_{\mathbf{u}}]$ in the FE system allows the party (or parties) holding the correct attributes to decrypt, with decryption in the FE system as $Enc(pk, (\varphi, m))$ [2]. The construction of key-policy ABE from FE is similar.

Note that, however, ABE fundamentally differs from FE: in ABE, the final goal is to allow or disallow access to a message, whereas in FE, the primary goal is to allow one to learn only a function of a message [61].

Many types of ABE exist. One example is the threshold-policy access control

(also known as Fuzzy IBE), a combination of ciphertext-policy access control and key-policy access control, in which the attributes label both the ciphertext and the key, and decryption succeeds if enough attributes in the ciphertext and key overlap [5, 13]. Another example is the non-monotonic ABE, which allows parties to efficiently express a negative attribute, thereby excluding certain parties from being authorized for decryption [5, 62]. Revocable ABE allows revocation of malicious parties or expired attributes from the system [5, 63]. Key delegation enables a party to create secret keys for other parties that own subsets of its attributes [5, 25]. Finally, Hierarchical ABE (HIBE) follows a similar idea to HIBE, utilizing a hierarchical structure and being particularly helpful when the number of attributes in the system is large.

Other properties, such as dynamicity or multi-authority, have been and will be further discussed in the general context of MPFE (see Sections 5 and 9). We only mention here the recent result of [64], which introduces multi-authority registered ABE: multiple (independent) key curators each manage their individual sets of attributes, thereby eliminating the limitation of centralized (or registered) ABE, where a single entity manages all the attributes in the system. This enhances the notion of registered ABE, firstly introduced in [65].

More information on ABE is available in [5, 66–68].

7.4 Order-Preserving Encryption (OPE)

Order-Preserving Encryption (OPE) is a symmetric encryption method that preserves the order of plaintexts in the ciphertexts. This means that if $m_1 < m_2$, then $Enc(k, m_1) < Enc(k, m_2)$, for any symmetric key k and any two messages m_1 and m_2 . Multi-input symmetric-key FE can filter by a given threshold, such as OPE. More generally, they can model any property preserved by Property-Preserving Encryption (PPE) [3, 23].

Suppose that the ciphertexts $c_i = Enc(pk, m_i)$ are stored on a server. For a client to retrieve all encrypted data with a corresponding plaintext value less than a given threshold t , the client sends $c = Enc(pk, t)$ and the secret key $sk[f]$ with f the binary comparison function: $f(m, t) = 1$, if $m < t$, and 0 otherwise. This allows the client to retrieve the ciphertexts that correspond to plain values below the threshold without exposing other information, such as the relative order of the items [3, 23].

A similar notion, Order-Revealing Encryption (ORE), reveals the order of two or more plaintexts without decryption, thus, without holding the decryption key. Both OPE and ORE find their utility in computation over encrypted data, for example, queries over encrypted databases. The paper [69] proposes a construction for ORE using MIFE. In [70], the authors introduce the notion of function-revealing encryption as a generalization of ORE to any multi-input function, as well as a relaxation of MIFE.

7.5 Predicate Encryption (PE)

In a Predicate Encryption (PE) scheme, the ciphertext is associated with a set of attributes **att**, called an index, and the decryption key is dependent on a predicate P . The decryption takes place correctly if and only if the set of attributes **att** satisfies the predicate P , that is, $P(\mathbf{att}) = 1$ [5, 71]. The set of attributes can be hidden or not, in which case we refer to attribute hiding PE (or private-index PE) and public-index, respectively. Both IBE and ABE schemes are predicate encryption systems with a public index [5]. In fact, anonymous IBE can be constructed from Inner Product PE (IPPE), a specific type of PE that supports the predicate of inner products (over $\mathbf{Z}_N$, for large N) [5, 71].

More information on PE and its applicability can be found in [5, 6].

7.6 Searchable Encryption (SE)

Searchable Encryption (SE) enables searches on encrypted data without decryption, thus minimizing the compromise of data confidentiality [36, 72]. Practical techniques for studying practical SE have been continuously developed since its introduction in [73]. The requirements and formal security definitions have been refined over time [36], including the necessary properties in practical applications, such as forward and backward privacy [74].

SE is closely related to other notions presented before. For example, the recent construction FEASE [75] is based on an anonymous KP-ABE. PE and SE overlap. For example, the Hidden Vector Encryption (HVE) introduced in [76] is both a case of PE and a subclass of SE.

More information on SE and its applicability can be found in [5, 6, 36, 72].

7.7 Fully Homomorphic Encryption (FHE)

Fully Homomorphic Encryption (FHE) is a strong cryptographic concept. Given a public key pk and the encryption of some messages $m_1, \dots, m_t$ under pk , and a description of a function f , one can construct an encryption of the message $f(m_1, \dots, m_t)$. If f is *univariate* (i.e., $f(m)$ is defined in a single variable), then we talk about *univariate FHE* [2], which seems closer to the definition of FE (rather than MPFE). Both FE and FHE perform some operation on the ciphertexts, but for FE, the result is in plaintext (a function applied to the plaintext), while for FHE, the result is encrypted (a function applied to the ciphertext) [21, 77]. FHE is not a form of FE, as the data is never decrypted. FE and FHE are distinct concepts that find utility in different scenarios.

Contrary to FE, FHE is a more studied concept, with more settled results in both theory and practice. The relation between FE and FHE has also been investigated in the literature [2, 78]. In particular, [79] shows how to build a succinct single-key FE scheme for general functions as a reduction from ABE and FHE. Under certain assumptions and specific cases, FE can imply FHE [78].

More information on FHE can be found in [5, 6, 80–82].

7.8 Others

The zoo of MPFE-related concepts is exceptionally vast and can not be extensively discussed here.

For example, FE is analogous to Secure Computation (SC), with the difference that functional encryption is completely non-interactive once a recipient obtains the secret key [2]. Instead, secure computation is an interactive protocol that involves communication between parties. Naturally, Secure Multi-Party Computation (SMPC), which extends SC to the multi-party case, is more closely related to MPFE. SMPC is a well-studied field in the scientific literature. More information on SMPC and its applicability can be found in [5, 83, 84].

The concept of obfuscation is closely related to the security study of MPFE, as it was considered from the first papers that introduced MIFE. In [22], Goldwasser et al. present constructions for a large class of MIFE, indistinguishably secure, under the assumption of indistinguishability obfuscation for circuits. Further, other papers, including very recent research, investigated the relation of (MP)FE with obfuscation [85–87].

Private Set Intersection (PSI) is a cryptographic mechanism that allows one to calculate the intersection of sets belonging to different parties without revealing anything except the intersection. FE can help build PSI; examples include [88, 89]. In particular, [88] introduces the notion of functional encryption for set intersection in the multi-client setting. Notably, PSI can also be related to FHE. For example, [90] shows that FHE can be utilized to construct PSI under specific settings. This again shows the strong bond and interconnectivity of cryptographic concepts.

Secret sharing (SS) is a cryptographic protocol that allows a dealer to share a secret among multiple parties, so that only an authorized set of parties can reconstruct it. The paper [25] shows that MAFE implies (computational) broadcast non-interactive Functional Secret Sharing (FSS). By functional secret sharing, as defined in [91], we refer to the generalization of secret sharing in which the parties can compute $f(s)$ for a function f in a family of functions, rather than simply computing the secret s .

8 Applicability

MPFE provides a non-interactive solution for a variety of applications, including those discussed further. Note that many use-cases overlap, and a strict classification in terms of functionality or field of interest is challenging.

8.1 Delegation of computation

The delegation of computation has been widely considered in the literature, in both single and multi-party settings [2, 3, 5, 6, 22, 92, 93].

8.1.1 Spam filtering on encrypted mail

A first examination of delegation of computation involves delegating spam filtering to a proxy server [2, 22]. FE could accommodate a proxy that filters out encrypted mail using specific criteria or a policy as follows [2]. The criteria for marking mail as spam or not-spam are modeled by a function f . The proxy is given the decryption key $sk[f]$ so that, at decryption, the proxy outputs 1 if m is classified as spam and 0, otherwise. Thus, the proxy verifies if the encrypted mail c is spam ($Dec(sk[f], c) = 1$) or not ($Dec(sk[f], c) = 0$) only by looking at its encryption and finds nothing about the content of the message itself [2].

8.1.2 Forwarding and routing

A generalization of the previous scenario is straightforward [2]. The same idea can be used to send emails based on specific criteria (e.g., forwarding important emails) or to route traffic in a network (e.g., by a router that selects the next hop for a received encrypted packet). In multi-input settings, MIFE can accept multiple inputs $m_1, \dots, m_n$ as the packet information (e.g., the source and destination IP addresses), the network topology (e.g., the edges [22]), and the network state (e.g., the overload of different hops or routes).

In all cases, the computation rule (modeled by f) receives as inputs $m_1, \dots, m_n$, which correspond to the ciphertexts $c_1, \dots, c_n$ used by the party delegated to perform the computation. The above scenarios also fit into the case of computing over an encrypted data stream [22].

8.1.3 Analytics

There are two main requirements for a scenario in which multiple parties $P_1, \dots, P_n$ want to delegate the computation of a function $f(x_1, \dots, x_n)$ on their data to a server S : (1) efficiency - the computation on each side P_i should be independent of n and (2) security of the server - a malicious server should not be able to convince the honest parties of an incorrect output of the computation [22].

MIFE allows the creation of analytics dashboards [77], also by integrating data originating from multiple companies. Following the idea presented in the previous paragraph, companies can encrypt their data and enable a third party, such as a cloud provider, to publish a dashboard based on predefined criteria, such as periodic key performance indicators [77]. While FE allows the creation of dashboards for a single data owner, MIFE opens the possibilities for multiple distinct data owners.

The applicability of the scenario is vast. Consider, for example, the n senders not being companies, but sensors in an IoT network or network monitors. They collect data and report to a central server in encrypted form, allowing the server to compute some functions on the reported results. This becomes increasingly important when it stands against some of the entities (possibly including the server) that are compromised. In this case, a security requirement is to keep the data sent by any honest sender private (naturally, up to the aggregated intended result). Research shows that MIFE can achieve this [23].

8.2 Computation over encrypted data

We further consider the case of computation over encrypted data. Note that there is an overlap with the use-case scenarios discussed in the previous subsection, as they also require computation over encrypted data. However, we now focus on applicability that is not centered on delegation.

8.2.1 Computation over encrypted data stream

MIFE finds its utility in law enforcement [22]. The ciphertexts $c_1, \dots, c_n$ can correspond to encrypted data streams such as voice or video calls. Law enforcement agencies may ask to run verification algorithms for suspicious behavior without accessing plaintext data $m_1, \dots, m_n$. Then, using MIFE, they can compute the result of $f(m_1, \dots, m_n)$ that evaluates the suspicious risk of the given conversations while not invading the privacy of the users [22].

Similarly, FE can be used in surveillance to identify a person within a set of encrypted images, without revealing anything else about the content of the images [77]. The same applies to computation over data streams originating from IoT devices, such as surveillance cameras. In this case, algorithms searching for suspicious activity could run directly on the encrypted data stream provided by the cameras [22]. As a particular case, image recognition algorithms could benefit from FE. For example, using facial recognition on encrypted images could filter encrypted images containing specific objects or persons [2].

Note that FE can be used in the streaming scenario if the whole data is encrypted in one shot and by a single party, while MIFE naturally extends to a streaming setting where the client does not hold all the data at one time, or/and there are multiple clients involved [23].

8.2.2 Private data release

MIFE can also be helpful in healthcare, accomodating data aggregation among hospitals that each own their own data [22]. Naturally, medical data are protected by patients' rights, which do not allow data to be made public for clinical trials or research under normal conditions. In MIFE, each hospital can encrypt its data, resulting in multiple ciphertexts $c_1, \dots, c_n$. Naturally, each hospital enforces its own security measures, so encryption is performed under distinct keys, also to avoid leakage among hospitals themselves (nevertheless, all keys are related to the master key msk). The hospitals could then agree on allowing researchers to compute aggregate functions $f(m_1, \dots, m_n)$ on their data, by simply handing the researcher a corresponding secret key $sk[f]$.

A second example is the controlled release of data, for instance, in the case of law enforcement. Let us consider the case of a bank that maintains encrypted financial records of its customers, so that some part of these data can be revealed to officials in case of law enforcement [25]. A specific example is the total value of cash owned by a customer or its interests, which must be made public to the tax office. To allow some authorities, such as the tax office, to compute authorized functions f on clients, the bank could issue secret keys $sk[f]$.

8.2.3 Multi-client data aggregation

MIFE (in particular, with f functioning as an inner product) is helpful in calculating weighted scores on sensitive data [94]. A specific case involves a manager who needs to evaluate the score of an entire team based on specific weights and the individual scores of the employees from the last evaluation. The goal is to determine the best team for a particular task. Similarly, it can be helpful to assemble a complex structure from parts that achieve a reasonable trade-off between quality and price. Each part is rated according to different quality characteristics and price, which are private, proprietary information of the manufacturer until the contract is signed [94].

8.2.4 Database query

FE allows fine-grained access to datasets by allowing a specific set of queries for different parties [2]. If the content of a dataset is encrypted using FE, a party can be allowed to query specific functions f in the data, and restricted to any other queries, simply by granting the corresponding decryption key $sk[f]$. Although FE is a tool for mining encrypted datasets, data originating from multiple sources might become short.

MPFE overcomes this and allows the extraction of aggregate information from several different data sources [3, 22]. A solution for a party P_i that is allowed to perform a class of specific queries on an encrypted database using ordinary FE would mean giving P_i a secret key $sk[f]$ for every authorized query. This implies a large, possibly unfeasible, set. In MPFE, in particular in MIFE, the solution can be sketched using a binary function $f(Q, m)$ that outputs the result of the query Q in the database m if Q is a valid query. Then P_i can encrypt the query Q in c_1 and use c_2 , the encrypted database m , to privately obtain the results of the query [22]. If the database is not encrypted as a whole, but rather each entry is individually encrypted, then the function f should be defined on all the entries, i.e., $f(Q, m_1, \dots, m_n)$. This fits the dynamic scenario where entries are frequently added, deleted, and updated, and re-encrypting the entire database is consuming. However, this does not scale well for large values of n [3, 22], and specialized solutions for SE (see Section 7) are currently better suited for these types of interrogations.

8.2.5 FE VPN

FE could be a solution to traffic inspection while avoiding the exposure of data by decryption, even for a short time, as is currently the case with TLS inspection [77]. An FE-VPN could allow deep packet inspection without fully decrypting the data, thus enabling users to detect malware or other threats without exposing the data in clear [77]. We are unaware of any such implementation today.

8.3 Machine Learning (ML)

MPFE finds its applicability in Machine Learning (ML). One use case is prediction on healthcare data. In [7], the authors exemplify with the case of a hospital that outsources the prediction of cardiovascular diseases to a Machine Learning as a Service (MLaaS) company, which should predict the correct results while preserving the privacy of patient data. Contrary to FHE, if FE is used, then the result is already decrypted, avoiding additional operations (the decryption itself and the sharing of the decryption key) [7]. The exact use case - prediction of cardiovascular diseases - is also previously discussed in [95]. As a natural generalization of this use case, MPFE nicely solves the case when multiple hospitals aggregate their data together in the training phase. In fact, according to [96], all existing Privacy-Preserving Federated Learning (PPFL) schemes that use FE employ an MIFE, or, more generally, an MPFE. According to other works too, MCFE shows its usability in Federated Learning (FL) [97, 98], including everyday scenarios, such as generating heatmaps for traffic density at underground stations using encrypted data to protect travelers' privacy [95].

8.4 Applications of ABE and IBE

We mention ABE and IBE separately because they are probably the particular cases of FE most applied in the real world. ABE integrates encryption and access control, being appropriate for sharing sensitive data within groups [5, 61, 77]. ABE finds its utility in securing digital wallets [77], healthcare [99], IoT [68, 100], and cloud [67]. In [101], the authors propose to use ABE - based on lightweight ECC - to secure Message Queue Telemetry Transport (MQTT) and MQTT-SN (for sensor networks), protocols used for Device-to-Device (D2D) communication in IoT. A survey of how IBE is used in vehicular ad-hoc networks is available at [102]. The paper [103] discusses identity-based authentication in VANETs. We discuss ABE and IBE with respect to standardization in Section 9. For more examples of applicability for ABE and IBE, including areas such as networking, healthcare, blockchain, and smart cities, the reader can refer to [6].

8.5 Others

The paper [104] introduces SACfe, an attribute-based functional encryption that can be used in online biometric authentication and access control.

In [105], a mechanism for detecting money laundering based on MCFE is proposed. This exemplifies the necessity of using labels, and thus the utility of MCFE upon MIFE. The main idea is based on a previous protocol to detect suspicious activity by modeling the system as a graph where the accounts are the vertices, the transactions are the edges, and the weights represent the money transferred. A two-client inner-product FE is further used to calculate similarity among the nodes.

Other applications include ordering encrypted data on the server side using FE [106], tax calculation [107], searchable encryption on multiple key-

words [108], data classification [109], applicability in parental control or smart cities [6].

9 Discussion

We discuss the state-of-the-art in (MP)FE, highlighting challenges and open problems related to different aspects, including security, efficiency, scalability, emergent domains, and post-quantum security. We also put a focus on decentralization and dynamism. However, an in-depth discussion can be much too technical for our purpose here. For the interested reader, the state of the art is nicely discussed in recent works [40, 41, 110–112]. We also discuss the presence of (MP)FE in the standardization process and existing specialized software.

9.1 Advances and challenges

9.1.1 Constructions

MPFE comes in many flavors, each with its own particularities. Some of these have been discussed in Section 6, but the list is not exhaustive. For example, the paper that introduced the notion of MIFE addressed the possibility that each party encrypts a single plaintext at every time step, which changes the security definitions and accepts a construction with succinct ciphertexts, dependent on the security parameter only [3]. In general, MPFE introduces a new way to look at FE, opening up many new possibilities for constructing a wide variety of properties [4]. Properties currently considered for specific implementations, such as revocation from ABE, could be extended to general FE. In addition, mechanisms such as traceability, which are expensive in current schemes [5], could be investigated further.

A great challenge in FE is to create secure, efficient, and scalable systems that accept a large possible family of functions for f [2]. Ideally, f can model any possible functionality.

For the particular case of ABE, in public index settings (see Section 6), ABE schemes to support essentially any access policy were defined, also achieving post-quantum security [113]. In the non-public settings, the state of the art is more limited.

Currently, most of the work has been performed for the inner product functionality [94], where the function f is defined by a collection of n vectors $\mathbf{y}_1, \dots, \mathbf{y}_n$, takes as input n vectors $\mathbf{x}_1, \dots, \mathbf{x}_n$ and outputs $f_{\mathbf{y}_1, \dots, \mathbf{y}_n}(\mathbf{x}_1, \dots, \mathbf{x}_n) = \sum_{i=1}^n \langle \mathbf{x}_i, \mathbf{y}_i \rangle$. These schemes are also known as Inner Product FE (IPFE). Refer to [37] as a standard reference for IPFE, as being the first paper to introduce efficient schemes for the IP functionality, based on a standard assumption (DDH). Besides IPFE, quadratic FE were also proposed, primarily based on pairings. Lately, the Attribute-Weighted Sums (AWS) has gained some attention, as a close generalization of the Inner Product (IP) that assigns distinct weights to different factors. Going beyond this, inspecting new cryptographic

approaches for general MPFE constructions, remains a field to explore. In particular, [114] claims to construct a private-key MCFE scheme for constant-arity functions that remains secure for a bounded number of key and encryption queries under the only minimal assumption that one-way functions exist.

Very recently, [110] has advanced the MCFE for attribute-based inner product functions in unbounded settings. More precisely, the paper demonstrates the applicability of function-hiding security of IPFE in MIFE capable of encoding unbounded-length vectors both at key generation and encryption. This overcomes previous limitations, where, for example, participants could only encrypt fixed-length data [110]. Another notable example is the work of [111], which introduces several results, including the first DDFE for inner products, secure under the LWE assumption in the standard model, as well as the first adaptively secure DDFE schemes for inner products in the stronger function-hiding setting (the schemes hide not only the plaintext, but also the function). In [111], the authors present the first construction of a DDFE for the more general AWS functionality with attribute-based access control, with some limitations.

9.1.2 Decentralization and dynamicity

The trusted authority in FE introduces a single point of trust and failure, while also creating a performance bottleneck. Also, it is implausible in scenarios involving multiple trust domains or organizations, where no one trusts the other to act as the trusted authority [2]. This is, for example, the case of private data release in the healthcare scenario (see Section 8), where hospitals should agree on who is taking the role of the authority. If all hospitals are located in the same region or country, local or national authorities could decide this, but the cross-border scenario imposes limitations. Works on decentralization have been discussed earlier in this paper.

Contrary to decentralization, much less work is performed for fully dynamic settings. Much of the existing work on MPFE is in static settings, assuming an initial setup with coordination among the parties [4]. Some works, such as ad-hoc MIFE [35] and DDFE [27], tackle the problem of dynamic FE with support for multiple users. Thus, the study of practical dynamical constructions remains a problem of interest.

New advances are made, with direct applicability to specific scenarios. An example is the advances in Streaming Functional Encryption (SFE) [112], a primitive that was recently defined in [115]. SFE permits computation on the data in streaming mode (see Section 8.2), at the time data become available, and in time proportional to just the size of the newly arrived data [112].

9.1.3 Security

Another challenge, commonly encountered in the construction of FE systems, is the prevention of *collusion attacks*. More players $P_1, \dots, P_k$ can combine their knowledge (including their secret keys $sk[f_1], \dots, sk[f_k]$), for example, to create new secret keys. In ABE, for instance, colluding parties can easily combine their

attributes such that their union satisfies the access policy [2]. Let us consider an access policy that only grants decryption rights to anyone who "is a student" and "is in the second year" and "studies engineering". Then P_1 , a second-year student in the arts, and P_2 , a first-year student in engineering, can collude to bypass the access policy. To overcome this, in ABE, the secret keys for a set of attributes (thus belonging to a single party) are bound by random values, such that the colluding parties cannot mix their keys [2]. In our example, all keys belonging to P_1 are blinded by a random value, and a different random value blinds all keys belonging to P_2 . Decryption only succeeds if it is performed with keys blinded using the same randomization [2].

Relatively recently, the authors of [116] list a significant number of vulnerable ABE schemes. They even found attacks against schemes introduced at top conferences, showing that even if the proposals have passed a strong peer review process and have been studied by many researchers, the schemes can still be vulnerable. As a consequence, the authors claim that formalizing strong attack models, which are easier to verify, simplifies the search for generic attacks and that simpler checking tools help identify vulnerabilities. As a particular note of interest in the decentralized settings, the authors discovered that mistakes are often made in transforming single-authority schemes into multi-authority schemes [116].

9.1.4 Quantum schemes and post-quantum resilience

Several IBE and ABE schemes are built on lattices [66, 117]. Other more general MPFE schemes are also built on lattices; refer to the existing surveys [5, 6] or recent advances [111] for examples.

The paper [52] extends FE to the quantum setting by providing definitions and a construction for a Quantum Functional Encryption (QFE) scheme, built upon quantum garbled circuits. The scheme allows for the evaluation of polynomial-sized circuits on arbitrary quantum messages. The paper also establishes that any indistinguishability-secure quantum MIFE scheme unconditionally implies the existence of quantum indistinguishability obfuscation [52], a result similar to the classic, non-quantum case [22]. Very recently, [40] defines the first DDFE for inner products whose security can be proven under the LWE assumption in the standard model. As a note, advances have been made in the construction of PE based on lattices too [118].

Future work is expected to explore further MPFE that withstand quantum attacks (post-quantum) or are defined in the quantum world (quantum protocols).

9.1.5 Efficiency and scalability

FE is still quite expressive. Typically, this is because some parameters of the scheme (e.g., the ciphertext size, the number of public key operations) scale with parameters such as the security level or the complexity of the family of accepted functions $\mathcal{F}$ [2, 22]. FE schemes are thus slower compared to other cryptographic

approaches (e.g., FHE, which has also received more attention from industry) that might be used, in some scenarios, for the same purpose. This becomes a problem when computations are performed on large datasets [7].

FE was also investigated in constrained environments, with [119] introducing the first lightweight FE scheme explicitly designed for resource-constrained devices, using Elliptic Curve Cryptography (ECC). The proposal has direct impact in IoT, in a scenario where devices collect and securely deliver data to a storage server to which an analyst can request access. The authors implement their proposal and run it on two ARM Cortex sensor devices, looking into both the execution time and energy consumption. The paper [98] proposes *PrivLDFL*, a lightweight and dynamic PPFL framework for resource-constrained devices and demonstrates its practicability on Raspberry Pi. Particularly, more research and experimentation have been done in the field of ABE running on constraint devices [6]. Very recently, improvements on compact IPFE were published. In [111], the authors propose compact constructions from several assumptions (including LWE-based, thus plausible quantum resistant) and propose an economic proof strategy to reuse the spaces to compact IPFE.

FE was also proposed to enhance security in MQTT [120]. Previously, other proposals only proposed ABE [101]. The authors of [120] propose a Proof-of-Concept (PoC) implementation using the *GoFE* library [121]. The solution aims to hide the message topic of a published message from the broker or other clients by having the broker evaluate a randomized FE ciphertext to obtain a boolean result indicating whether to forward it.

We place a complete comparative complexity table outside our current scope. Such an analysis depends directly on the construction of the schemes (including their functionalities, hardness assumptions, and other properties), and we do not aim to discuss schemes' constructions, but rather maintain the discussion at the conceptual level. For the common case of the IP, however, the ciphertext size typically remains $\mathcal{O}(l)$, where l is the vector length (same as in FE). As an exception, DDFE requires a ciphertext length of $\mathcal{O}(l + \log(n))$. While the Cosmian implementation (on BLS12-381 elliptic curve) [122] shows some difference in decryption time for DMCFE compared to MCFE, the main bottleneck in dynamic settings remains the setup phase. Research also shows that HW/SW codesign acceleration speedups 2.3–20.0 times compared to the optimized GoFE library for quadratic functions [123].

9.1.6 Emerging domains

We further discuss the recently published utility of MPFE in emerging domains such as ML, neural networks, or federated analytics.

Lately, FE has found its utility in Privacy Preserving Machine Learning (PPML). However, it is not yet a well-established technique in the field, such as FHE or SMC. The paper [7] provides a systematization of PPML, summarizing the state-of-the-art literature, and also analyzing the performance and usability of the available FE libraries and their applications to PPML.

PPML deals with the problems related to data and model privacy during all

stages of ML, from training to deployment. Preserving privacy becomes even more challenging when ML is offered as a service, running, for example, in the cloud, with clients that are responsible for training [7]. Contrary to the case of FHE, where the server obtains encrypted predictions, in the case of FE, the server receives the results in plaintext by computing a function on the encrypted data. This might be appropriate for some applications only. There is also a need for PPML to efficiently train and predict on encrypted data, which FE can (to some extent) accommodate. MPFE-based schemes could allow decentralization and avoid a single authority, thus a single point of trust [7].

The two main types of FE can be applied in ML: (1) Inner-product FE-based - supports both training and prediction over encrypted data and (2) Quadratic FE-based - supports prediction over encrypted data, while training is performed over plain data. At the moment, quadratic FE is faster than inner-product FE in prediction [7].

According to the same paper [7], FE has demonstrated its functionality up to a 5-layer neural network, while complex neural networks such as VGGNet, AlexNet, and GoogleNet are considered far from implementing FE. Lower-layers neural networks have been considered previously in the literature in conjunction with FE (e.g., 2-layer neural network in [95]). The authors of [96] point out that existing FE-based PPFL schemes have faced several security issues because of the misuse of MIFE. To overcome this, they propose a new FE called dual-mode decentralized multi-client FE (2DMCFE) [96].

Advantages and disadvantages of using MCFE for federated analytics are nicely summarized in the introduction of [97]. On the benefits side, they remark that it withstands collusion between clients and the server, eliminates the need for a TTP, and offers the possibility to ensure well-formed ciphertexts. On the disadvantages side, the authors mention the communication overhead caused by frequent interactions, the lack of support for client dropout, and scalability issues [96].

9.1.7 Impact

The impact of FE results directly from the broad applicability discussed in Section 8. What initially aimed to expand the vision of encryption systems [21], has grown its significance in several ways.

The impact of ABE has not been particularly discussed in the paper, as our focus is on general MIFE. However, Waters give a nice three-fold impact of ABE over time [61]: (1) ABE as a stand-alone concept, e.g., sharing data according to access control policies (e.g., grant access control by location, using, among others, the GPS location as attribute - along with the identity of the driver - in an ABE scheme that encrypts sensitive data) (2) ABE as building block for other cryptographic systems, including garbled circuits, traitor tracing and non-interactive zero knowledge proofs, and (3) ABE as inspiration to rethink encryption in new ways, leading to FE. Starting with ABE, or using ABE as one of its main pillars, FE grew in impact in similar ways [4]: (1) helped to unify the existing literature on encryption systems (e.g., IBE, ABE), (2) helped to define

new types of encryption systems (e.g., partially hiding FE), and (3) served to understand the relationship with other systems and notions (e.g., obfuscation).

9.2 Standardization

FE, mostly in particular forms such as ABE or IBE, is present in standards. Table 6 provides a list of technical specifications, including references to FE and specifics such as ABE.

ETSI identified ABE as a key enabler technology for access control in distributed systems, including IoT and 5G [124]. ETSI TS 103 532 [125] specifies trust models, functions, and protocols that utilize ABE as a foundation for attribute-based access control, while ETSI TS 103 458 [126] discusses specific use cases in mobile, WLAN, and cloud environments for the utility of ABE in protecting personal data. ETSI TS 103 964 [127] discusses a proof method via a challenge-response authentication protocol based on CP-ABE. Other two documents refer to IBE in the classical [128] and quantum [128] models.

The National Institute of Science and Technology (NIST) defines Attribute-based Access Control (ABAC) and describes its functional components [129]. NIST SP 800-162 defines ABAC as an access control method in which rights are granted based on the assigned attributes of the subject and the object, among others (e.g., policies and environmental conditions such as time, day of the week, and location). Although ABAC is not directly related to any underlying mechanism, ABE is a good candidate for implementing ABAC [77]. In fact, years later, NIST IR 8450 delves into how encryption, particularly IBE and ABE, can be incorporated into access control mechanisms.

We are aware of no standard or specification referring to the general notion of MPFE. Nevertheless, the mentioned documents touch similar notions for ABE, such as multi-authority ABE for decentralization [130].

9.3 Libraries and implementations

GoFE (in Go) [121] and *CiFEr* (in C) are two libraries dedicated to FE. They both appeared as a result of a dedicated EU project called FENTEC [138]. Although not updated for several years now, they implement FE schemes for both linear (inner products) and quadratic polynomials (*GoFE* only). *GoFE* clearly outlines the invocation of *GoFE* functions from JavaScript [139] and the use of libraries in applications developed in other programming languages [140].

The paper [141] presents ABE libraries, identifying 11 libraries of interest. Their analysis highlights four central schemes, based on criteria such as maintenance, the implementation of broken schemes at the time of writing, underlying math libraries used, and design goal (production or research): *OpenABE* [142], *Rabe* [143], *Charm* [144], and *GoFE* [121], which we have already mentioned before.

Other libraries are available; however, dedicated implementations of MCFE that are up-to-date are minimal. *PyFE* [145] is a Python library for FE, not updated in the last 5 years. Other specific implementations of FE schemes

Table 6: **Specifications involving FE**

Focus	Ref.	Document
ABE	[126]	ETSI TS 103 458. CYBER; Application of Attribute Based Encryption (ABE) for Personally Identifiable Information (PII) and personal data protection on IoT devices, WLAN, cloud and mobile services - High level requirements
ABE	[125]	ETSI TS 1103 532 V1.2.1 (2021-05) - CYBER; Attribute Based Encryption for Attribute Based Access Control
ABE	[127]	ETSI TS 103 964 V1.1.1 (2025-02) - Cyber Security (CYBER); A Verifiable Credentials extension using Attribute-Based Encryption
IBE	[128]	ETSI TR 103 719 V1.1.1 (2022-03) - Guide to Identity-Based Cryptography
IBE	[131]	ETSI TR 103 618 V1.1.1 (2019-12) - CYBER; Quantum-Safe Identity-Based Encryption
ABE	[132]	ETSI TS 104 015 V1.1.1 (2025-02) - Cyber Security (CYBER); Quantum-Safe Cryptography (QSC); Efficient Quantum-Safe Hybrid Key Exchanges with Hidden Access Policies
IBE	[133]	ETSI TR 103 619 V1.1.1 (2020-07) - Migration strategies and recommendations to Quantum Safe schemes (Note: it only contains a note on IBE, as a particular case of FE)
IBE	[134]	ETSI TS 103 816-3 V1.1.1 (2021-07) - Publicly Available Specification (PAS); CYBER; Connecting Products based on MIKEY-SAKKE; Part 3: One-to-One Messaging (Note: references to IBE exist in Part 1 and 4 of the document too)
IBE	[135]	ETSI TR 103 570 V1.1.1 (2017-10) - CYBER; Quantum-Safe Key Exchanges
ABE	[136]	ETSI TS 103 485 V1.1.1 (2020-08) - CYBER; Mechanisms for privacy assurance and verification
ABE	[137]	ETSI TR 103 370 V1.1.1 (2019-01) - Practical introductory guide to Technical Standards for Privacy
IBE, ABE	[130]	NIST IR 8450 - Overview and Considerations of Access Control Based on Attribute Encryption
ABAC	[129]	NIST SP 800-162 - Guide to Attribute Based Access Control (ABAC) Definition and Considerations

are available, for example Cosmian [122], which implements three FE schemes, including MCFE schemes. *PyMIFE* [146] is a Python library dedicated to MIFE, but rather a personally developed library, which can only be used for simple testing purposes. Older and unmaintained libraries also include [147]. *sumFE_v2* [148] is a dedicated implementation for lightweight FE for IoT. Finally, we mention *sumFE* [149], the implementation from [119]. Other general-purpose libraries, such as FLINT [150] or PBC [151], can be used for FE purposes, as noted in [7].

Some papers on FE for ML analyze the existing FE libraries and their utility [7, 95].

To overcome the complexity and impracticability of existing FE at the time, *Iron* - a practical general-purpose MIFE implementation for Intel SGX - was proposed [152]. *Iron*'s security relies on trust in the manufacturing process and the robustness of Intel SGX, which utilizes hardware support for isolated program execution environments (enclaves). Currently, there is no hardware acceleration support for FE implementation [7]. Such implementations of (MC)FE on GPUs or FPGAs, should be investigated.

There is a lack of MCFE practical implementations, open-source solutions, and production-ready options. Compared to other similar cryptographic protocols, such as FHE, for which stable libraries are available, the gap is considerably larger.

9.4 Limitations

At the moment of writing, MPFE has not reached its full maturity and still faces many limitations. Although efficient MPFE schemes for simple functions are feasible, MPFE for arbitrary functions is computationally heavy and impractical in the real world. Although dynamic MPFE has been defined in theory, most current implementations are static and impose a heavy burden on an expensive setup phase, requiring all participants to be online simultaneously. Switching to a decentralized setup is affordable for simpler functionalities (e.g., weighted sums), but remains unaffordable for arbitrary functionalities. As a result of these limitations, there is a lack of production-ready source code for implementing general MPFE, and there is a refusal to adapt it to standards, with ABE and IBE being the notions used in real-world implementations.

10 Conclusion

MCFE has great potential and promises a large area of applicability. The field is relatively new and still not fully mature, leaving space for investigation and improvement. The current paper revises the main definitions, properties, and applicability scenarios. We also highlight the common tendency of including older and more established notions, such as ABE, in technical specifications from organizations and institutes, while there is a notable lack of references to newer concepts. The present study reveals a lack of efficient MCFE schemes

that can be utilized on a large scale or are mature enough for use in production environments, highlighting the need for open-source and professional implementations.

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